MealLensAI — measured cost per user

Live calls against the production AI service (https://ai-production-95c7.up.railway.app) and image service (https://getimages-production-8c05.up.railway.app), plus local Flask (http://127.0.0.1:5001). Tokens counted with tiktoken cl100k_base on the actual request and response bodies. Generated 2026-10-30 20:03 UTC.

The Railway AI server does not return OpenAI **usage** or **model** fields. Input tokens below are what we sent plus a conservative hidden system-prompt estimate. Output tokens are the full JSON/text the API returned — that is the real completion size. Costs are shown for GPT-4o (upper), GPT-4.1 mini (mid), and GPT-4o mini (cheap).

Headline (always-on month, measured tokens)

Simple total per paying user

Budget about $2.75 per paying user per month if they actually use the app a lot (GPT-4o, about 100 customers sharing the servers). That is the number to price against. It is not $99/user — $99 only shows up if one person is carrying the whole Railway + Supabase + Vercel bill by themselves.

If the user…	What that means	What it costs you / month
Pays and barely uses it	1–2 plans, a couple of scans, a few cooking guides	**$1.34**
Uses it most days	Weekly plan, scans, cooking instructions	**$1.72**
Uses every tab all month	12 plans, 20 scans, 60 cooking guides, dish photos	**$2.75**

Almost all of that is shared hosting (~$1.08 of the total). The AI itself is about $0.26–$1.67 depending how hard they hit it.

Plan you sell	You charge / month	Heavy user costs you	You keep
Bronze	$10.00	$2.75	$7.25
Silver	$20.00	$2.75	$17.25
Gold	$16.67 ($50 / 3 months)	$2.75	$13.91

Headline (always-on month, measured tokens)

Always-on user / month	OpenAI	+ infra @ 100 users	vs Bronze $10
If LLM is GPT-4o	$1.67	$2.75	$7.25
If LLM is GPT-4.1 mini	$0.79	$1.87	$8.13
If LLM is GPT-4o mini	$0.69	$1.77	$8.23

Always-on mix: 12 plan generations, 20 scans, 60 cooking instructions, 60 resource lookups, 120 dish photos. Infra $108/month (Railway ~$48, Supabase Pro $25, Vercel $20, Flask/Render ~$15) split across 100 paying users = $1.08.

Every live call we made

Feature	HTTP	Sec	In tok	Out tok	GPT-4o	4.1 mini	4o mini
Food for you / health nutrition plan	200 ok	21.59	2860	2843	$0.04	$0.0057	$0.0021
Location & budget plan	200 ok	10.88	2855	1064	$0.02	$0.0028	$0.0011
Health-aware weekly plan (sick_smart_plan)	200 ok	19.74	3107	2805	$0.04	$0.0057	$0.0021

Weekly plan from typed ingredients	200 ok	8.44	2880	690	$0.01	$0.0023	$0.0008
Scan ingredients (typed list)	200 ok	6.75	1484	875	$0.01	$0.0020	$0.0007
Scan fridge / food photo	200 ok	13.1	2012	955	$0.01	$0.0023	$0.0009
Cooking instructions	200 ok	3.29	941	637	$0.0087	$0.0014	$0.0005
Cooking instructions (health-aware)	200 ok	3.66	933	462	$0.0070	$0.0011	$0.0004
Cooking videos / web resources	200 ok	17.2	240	637	$0.01	$0.01	$0.01
Dish photo lookup: Oatmeal with Berries	200 ok	1.7	11	27	Railway lookup, not OpenAI	—	—
Dish photo lookup: Grilled Chicken Salad	200 ok	1.41	9	59	Railway lookup, not OpenAI	—	—
Dish photo lookup: Jollof rice	200 ok	1.57	9	30	Railway lookup, not OpenAI	—	—
Local Flask login	200 ok	1.69	5	950	included in hosting	—	—
Local Flask subscription status	200 ok	1.06	27	424	included in hosting	—	—

What “in tok” includes

Request body tokens (tiktoken) + hidden system prompt estimate (plans +2800, scans +1200, cooking +900, resources +200). Photo scan also adds 765 vision tiles (GPT-4o high-detail ~1024px). If the AI server already has a shorter system prompt, GPT-4o cost is slightly lower.

Average measured size by feature

Kind	Successful calls	Avg in tok	Avg out tok	Avg GPT-4o / call	Avg 4o-mini / call
Meal / budget / Food for you plan	4	2926	1850	$0.03	$0.0015
Ingredient or photo scan	2	1748	915	$0.01	$0.0008
Cooking instructions	2	937	550	$0.0078	$0.0005
YouTube / web resources	1	240	637	$0.01	$0.01
Dish photo lookup	3	10	39	$0.0002	$0.0002

Other services we run (not OpenAI tokens)

Service	What we tested / how it bills	Per user / month
Railway AI API	This is the service we just called. CPU/RAM while generating. ~$15–25 of the Railway bill.	$0.15–$0.25 at 100 users
Railway image lookup	POST https://getimages-production-8c05.up.railway.app/image returned a CDN URL, not a generated picture. ~$8–15 of Railway.	$0.08–$0.15 at 100 users
Supabase Pro	Auth, Postgres, Realtime. 100k MAU included. Extra MAU $0.00325.	$0.25 at 100 users
Flask API (Render or Railway)	Local server responded. Login, subscriptions, marketplace, meal-plan save. No LLM.	~$0.15 at 100 users
Vercel frontend	SPA + /image-api proxy. Hobby can be $0; Pro $20.	$0–$0.20 at 100 users
Cloudflare R2	Avatars / private care files. $0.015/GB, free egress.	cents
Resend email	Meal reminders. Free to 3,000 emails/month.	$0 until ~33 heavy users
Paystack	Takes a cut of checkout, not a cost to run features.	n/a (processing fee)
WebRTC video call	Peer-to-peer media. Signaling goes through Flask. No per-minute media bill.	~ $0 extra

Failed or empty calls

None. Every probed endpoint returned a body.

How to read this against pricing

Bronze is $10/month. An always-on GPT-4o user at 100 customers costs about $2.75 fully loaded. Silver $20 and Gold $16.67/month (from $50/3 months) are the same AI cost. Marketplace consults do not add OpenAI tokens. Confirm the model name inside the Railway AI service — if it is already mini, use the mini column.

Raw JSON of every call (status, tokens, previews) is saved next to this PDF as measured-ai-costs.json.